Does the Age at Which a Shock Hits Matter?

Violence from before birth to school age, using the end of paramilitary violence in Colombia

Idea 1: statement and assessment

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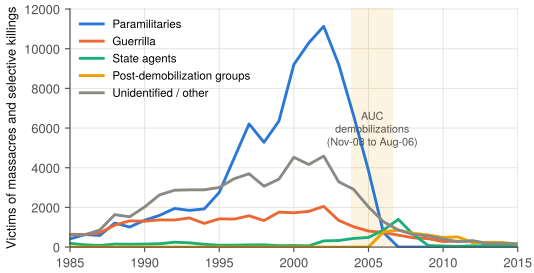
Question

Does the long-run effect of a shock depend on the age at which the child receives it: before birth, in infancy, in preschool, at school age?

- The sensitive-periods view says yes (Cunha and Heckman, 2007, *AER*; Almond, Currie and Duque, 2018, *JEL*).
- The evidence is thick for the prenatal period (famine, pollution, stress) and thin after birth: age profiles are estimated by comparing cohorts hit at different ages by a shock whose timing is rarely exogenous, with people located where they are observed, not where they were hit.
- Violence is the natural shock for this question: large, frequent, well dated. The existing profiles (Burundi, Nigeria, Peru, Germany, Colombia) all use the arrival or intensity of violence and share those two weaknesses.

Setting: the end of violence instead of its arrival



- Paramilitary blocs demobilize one by one, 2003–2006; zero victims after 2007. The end month T_m is observed for each municipality.
- Within a municipality, age at T_m traces the whole profile: born after (never exposed), just before (in utero only), 1–2 years before (infancy), 3–5 (preschool), 6–11 (school age).
- The end date comes from a national negotiation, not from local trends.

Specification

i born in municipality m, cohort c; $k = c - T_m$ is age at the end.

$$Y_{imc} = \sum_{k \neq -1} \beta_k \, \mathbf{1}[c - T_m = k] + \mu_m + \lambda_{c \times d(m)} + \varepsilon_{imc}$$

- β_k is the effect of exposure up to age k, relative to the never-exposed cohort of the same municipality, net of department-by-cohort trends. The profile $\beta_0, \beta_1, \ldots, \beta_{11}$ is the object.
- Staggered timing: Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024). Validity: pre-conception placebos; Rambachan and Roth (2023) bounds.
- Municipality of birth, individual microdata: GEIH at 20–36 (schooling, employment, earnings); Saber 11 at 17.

What it would add

- **A different source of variation.** Age at the *end* of a shock, with the end set by a national negotiation. It addresses the two weaknesses at once: timing not driven by local trends, location fixed at birth.
- **The whole profile in one design:** prenatal only, infancy, preschool, school age, against never-exposed cohorts of the same municipality.
- **Mechanism:** profile by household displacement (migration module) and by maternal health at birth (vital statistics).

The question is not new. What would be new is a credibly identified postnatal profile.

Assessment

Already well covered. Bundervoet, Verwimp and Akresh (2009, *JHR*); Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*); León (2012, *JHR*); Duque (2024, *JCR*); Akbulut-Yuksel, Tekin and Turan (2025, *JHR*). Same question, same fixed-effects design.

Too little variation. 70% of end dates fall within eighteen months. In-utero-only is a nine-month birth window; never-exposed cohorts become adults in 2026; with current data they exist in 28 municipalities.

Verdict. The only novelty is the instrument, and the instrument is thin. Not a job market paper as it stands.

Appendix. Age at exposure to violence: what the literature finds (1/2)

Paper	Setting	Variation	Outcome	Effect by age
Bundervoet, Verwimp and Akresh (2009, *JHR*)	Burundi, 1993–98	province timing × cohort	height-for-age, children under 5	-0.047 SD per month of exposure
Chamarbagwala and Morán (2011, *JDE*)	Guatemala, 1960–96	department of birth × cohort, three war periods	years of schooling	rural Mayan men: -0.27, -0.71, -1.09 years by period; women: -0.12, -0.47, -1.17; school-age exposure
Shemyakina (2011, *JDE*)	Tajikistan, 1992–98	region × cohort	completion of mandatory schooling	girls of school age 12–13% less likely to complete†
Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*)	Nigeria, Biafra 1967–70	ethnicity × cohort	adult height	losses at every age from birth to adolescence; largest for adolescents
León (2012, *JHR*)	Peru, 1980–93	district of birth × cohort	years of schooling	per year of exposure: in utero -0.07, 0–3 -0.05, 4–6 -0.05, 7+ ≈ 0. Replication with full census (Huaroto, Gonzales and Pacheco, 2026, *JHR*): pre-conception effects significant; only in utero robust

Appendix. Age at exposure to violence: what the literature finds (2/2)

Paper	Setting	Variation	Outcome	Effect by age
Akbulut-Yuksel (2014, *JHR*)	Germany, WWII	city destruction × cohort	schooling, height, earnings	school-age cohorts: about -0.4 years, -1 cm, -6% earnings†
Minoiu and Shemyakina (2014, *JDE*)	Côte d'Ivoire, 2002–07	region × cohort, surveys before, during, after	height-for-age	negative for children in affected regions; channel: household economic losses
Singhal (2019, *JDE*)	Vietnam, 1965–75	bombing intensity, IV	adult mental distress	$+16$ pp severe distress, about 50% of the mean; early-life exposure
Duque (2024, *JCR*)	Colombia, 1980s	municipality-month × cohort, mother FE	dropout, Saber 11	higher homicide rate in early childhood: more dropout, lower scores; 1 SD ≈ -0.1 years†
Akbulut-Yuksel, Tekin and Turan (2025, *JHR*)	Germany, WWII	city bombing × cohort	mental health at 55–70	1 SD bombing: -10% mental-health index, $+16\%$ depression; concentrated under age 5 and in utero

† magnitude from a working-paper version. **Reading.** Every row: birth or residence location × cohort with fixed effects, variation from the arrival or intensity of violence. Profiles disagree: schooling damage largest before school in Peru and in Colombia, at school age in Germany and Guatemala; height damage largest in infancy in Burundi and in adolescence in Nigeria; mental health damage in early childhood in Germany. The question is crowded; the postnatal profile is not settled.